



MATHEMATICS

Topic:

Indefinite Integration

Date: 27/04/2026

MARKING SCHEME

For each question, you will be awarded 4 marks if you have darkened only the bubble corresponding to the correct answer and zero mark if no bubble is darkened. In case of bubbling of incorrect answer, minus one (-1) mark will be awarded.

1. $\int \sqrt{\sin x} \cos x \, dx =$

1) $\frac{2}{3} \sqrt{\sin x} + c$ 2) $\frac{2}{3} (\sin x)^{3/2} + C$

3) $\frac{2}{3} \sin^{2/3} x + c$ 4) $\sin^{3/2} x + c$

2. $\int \cos^3 x \cdot \sin x \, dx =$

1) $\frac{\cos^3 x}{4} + c$ 2) $-\frac{\cos^3 x}{3} + c$

3) $-\frac{\cos^4 x}{4} + c$ 4) $\sin x + c$

3. $\int \sqrt[3]{\sin x} \cos x \, dx =$

1) $\frac{4}{3} \sin^{\frac{4}{3}} x + c$ 2) $\frac{3}{4} \sin^{\frac{4}{3}} x + c$

3) $-\frac{3}{4} \sin^{\frac{4}{3}} x + c$ 4) $-\frac{4}{3} \sin^{\frac{4}{3}} x + c$

4. $\int \frac{\sec^2 x}{(1 + \tan x)^3} \, dx =$

1) $\frac{1}{2(1 + \tan x)^3} + c$ 2) $\frac{1}{(1 + \tan x)^3} + c$

3) $\frac{-1}{(1 + \tan x)^2} + c$ 4) $\frac{-1}{2(1 + \tan x)^2} + c$

5. $\int \frac{\cos 2x}{(\sin x + \cos x)^2} \, dx =$

1) $\log|\sin x + \cos x| + c$

2) $\log|\sin x - \cos x| + c$

3) $\frac{1}{2} \log|\sin x - \cos x| + c$

4) $\frac{1}{2} \log|\sin x + \cos x| + c$

6. $\int \frac{(\cos x)^{n-1}}{(\sin x)^{n+1}} \, dx =$

1) $\frac{-\cot^n x}{n} + c$

2) $\frac{-\cot^n x}{n+1} + c$

3) $\frac{\cot^n x}{n} + c$

4) $\frac{\cot^n x}{n+1} + c$

7. If $\frac{dI}{dY} = 3^{\cos y} \cdot \sin y$ then I is equal to

1) $3^{\cos y} + c$

2) $-\frac{3^{\cos y}}{\log 3} + c$

3) $\sin y + c$

4) $3^{\sin y} + c$

8. $\int (x^{7m} + x^{2m} + x^m)(2x^{6m} + 7x^m + 14)^{1/m} \, dx$ equal to

1) $\frac{(x^{7m} + x^{2m} + x^m)^{m+1}}{m+1} + c$

2) $\frac{1}{14(m+1)} (2x^{7m} + 7x^{2m} + 14x^m)^{\frac{m+1}{m}} + c$

3) $(2x^{7m} + 7x^{2m} + x^m)^{\frac{m+1}{m}} + c$

4) $\frac{1}{14(m+1)} (2x^{7m} + 7x^{2m} + 14x^m) + c$

9. $\int 3^{-\log_9 x^2} \, dx =$

1) $2 \log |x| + c$

2) $\log |x| + c$

3) $-\log |x| + c$

4) $-2 \log |x| + c$

10. $\int \frac{x}{\sqrt{x^2 + 7}} \, dx =$

1) $\sqrt{x^2 + 7} + c$

2) $2\sqrt{x^2 + 7} + c$

3) $\frac{2}{\sqrt{x^2 + 7}} + c$

4) $\sinh^{-1}\left(\frac{x}{\sqrt{7}}\right) + c$

11. $\int (\operatorname{Cosec}^{-1} x)^n \cdot \frac{1}{x\sqrt{x^2 - 1}} \, dx =$

1) $\frac{(\operatorname{Cosec}^{-1} x)}{n+1} + c$

2) $-\frac{(\operatorname{Cosec}^{-1} x)^{n+1}}{n+1} + c$

3) $\frac{1}{(\operatorname{Cosec}^{-1} x)^{n+1}} + c$

4) $\frac{-1}{(\operatorname{Cosec}^{-1} x)^{n+1}} + c$

12. $\int \frac{\cot x}{\log \sin x} dx =$

- 1) $\log(\log(\sin x)) + c$ 2) $\log(\log(\sec x)) + c$
3) $\log(\log(\cot x)) + c$ 4) $\log(\log(\operatorname{cosec} x)) + c$

13. $\int \sin 5x \cdot \cos x \, dx =$

- 1) $\frac{\sin 6x}{12} + \frac{\sin 4x}{4} + c$
2) $-\frac{\cos 6x}{12} - \frac{\cos 4x}{8} + c$
3) $\frac{-1}{12} [\cos 6x + 3 \cos 4x] + c$
4) $\frac{-\cos 5x \cdot \sin x}{5} + c$

14. $\int 2 \sin 5x \sin 3x \, dx =$

- 1) $\frac{\sin 2x}{4} - \frac{\sin 8x}{16} + c$
2) $\frac{1}{8} [4 \sin 2x - \sin 8x] + c$
3) $\frac{1}{16} [4 \cos 2x - \cos 8x] + c$
4) $\frac{1}{16} [4 \cos 2x + \cos 8x] + c$

15. $\int \cos x \cdot \cos 2x \, dx =$

- 1) $\frac{\sin 3x + 3 \sin x}{6} + c$ 2) $\frac{\sin 3x + \sin x}{6} + c$
3) $\frac{\sin 3x - 3 \sin x}{6} + c$ 4) $\frac{\cos 3x + 3 \cos x}{6} + c$

16. $\int \frac{\tan x - \tan \frac{x}{2}}{1 + \tan x \tan \frac{x}{2}} dx =$

- 1) $\log \left| \sec \frac{x}{2} \right| + c$ 2) $2 \log \left| \cos \frac{x}{2} \right| + c$
3) $2 \log \left| \sec \frac{x}{2} \right| + c$ 4) $\log \left| \cos \frac{x}{2} \right| + c$

17. $\int \frac{1}{\sqrt{1-4x^2}} dx =$

- 1) $\sin^{-1} 2x + c$ 2) $\frac{1}{2} \sin^{-1} 2x + c$
3) $(\sin^{-1} 2x)^2 + c$ 4) $\frac{1}{2} (\sin^{-1} 2x)^2 + c$

18. $\int \frac{1}{x\sqrt{x^2-16}} dx =$

- 1) $\frac{1}{4} \sec^{-1} \frac{x}{4} + c$ 2) $\sec^{-1} \frac{x}{4} + c$
3) $\frac{1}{4} \sec^{-1}(4x) + c$ 4) $\frac{1}{4} \operatorname{cosec}^{-1} \frac{x}{4} + c$

19. $\int \frac{3}{\sqrt{9z^2-1}} dz =$

- 1) $\cosh^{-1}(3z) + c$ 2) $\sinh^{-1}(3z) + c$
3) $\frac{1}{2} \cosh^{-1}(3z) + c$ 4) $\frac{1}{2} \sinh^{-1}(3z) + c$

20. $\int (\tan^{10} x + \tan^{12} x) dx =$

- 1) $\frac{\tan^9 x}{9} + c$ 2) $\frac{\tan^{13} x}{13} + c$
3) $\frac{\tan^{11} x}{11} + c$ 4) $\frac{\tan^8 x}{8} + c$

21. If $\int \frac{\sin^2 x - \sin^2 \alpha}{\cos x - \cos \alpha} dx = f(x) + Ax + B$,

then

- 1) $f(x) = 2 \sin x$, $A = \cos \alpha$
2) $f(x) = 2 \sin x$, $A = 2 \cos \alpha$
3) $f(x) = \sin x$, $A = \cos \alpha$
4) $f(x) = \sin x$, $A = 2 \cos \alpha$

22. $\int (1+5x+10x^2+10x^3+5x^4+x^5) dx$

$= \frac{(1+x)^p}{6} + c$ then p is

- 1) 5 2) 6 3) 7 4) 8

23. $\int \frac{1+x^5}{1+x} dx$ is equal to:

- 1) $1-x+x^2-x^3+x^4+c$
2) $x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}+c$
3) $(1+x)^5+c$ 4) $(1-x)^5+c$

24. $\int \frac{1}{1 + \cos ax} dx =$

1) $\frac{1}{a} (\operatorname{Cosec} ax + \cot ax) + c$

2) $\cot \frac{ax}{2} + c$ 3) $\tan \frac{ax}{2} + c$

4) $\frac{1}{a} (\operatorname{Cosec} ax - \cot ax) + c$

25. $\int \frac{|x|}{x} dx =$

1) $x + c$ 2) $-x + c$
3) $|x| + c$ 4) $x^2 + c$

26. $\int \cot^2 x dx =$

1) $-\cot x - x + c$ 2) $\tan x - x + c$
3) $\cot x - x + c$ 4) $\sin x - x + c$

27. $\int x\sqrt{x} dx =$

1) $\frac{3}{2} x^{3/2} + c$ 2) $\frac{2}{5} x^{5/2} + c$

3) $\frac{5}{2} x^{5/2} + c$ 4) $\frac{3}{2} \sqrt{x} + c$

28. $\int (\tan x - \cot x)^2 dx =$

1) $\tan x + x + c$ 2) $\tan x - x + c$
3) $\tan x - \cot x + c$ 4) $\tan x - \cot x - 4x + c$

29. $\int \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) dx =$

1) $\sin x + c$ 2) $-\sin x + c$
3) $\cos x + c$ 4) $-\cos x + c$

30. $\int \frac{ax+b}{cx+d} dx =$

1) $\frac{ax}{c} - \frac{bc-ad}{c^2} \log |cx+d| + c$

2) $\frac{ax}{c} + \frac{bc-ad}{c^2} \log |cx+d| + c$

3) $\log \left| \frac{ax+b}{cx+d} \right| + c$

4) $\frac{ax}{c} + \frac{b}{c} \log |cx+d| + c$

31. $\int \frac{(6x-5)}{2\sqrt{3x^2-5x+6}} dx =$

1) $\frac{1}{\sqrt{2}} \sqrt{3x^2-5x+6} + c$ 2) $\sqrt{3x^2-5x+6} + c$

3) $\frac{1}{2} \sqrt{3x^2-5x+6} + c$ 4) $\frac{1}{3} \sqrt{3x^2-5x+6} + c$

32. $\int \frac{dx}{(\arcsin x)^3 \sqrt{1-x^2}} =$

1) $\frac{-1}{2(\arcsin x)^2} + c$ 2) $\frac{1}{2(\arcsin x)^2} + c$

3) $\frac{1}{3(\arcsin x)^3} + c$ 4) $\frac{1}{4(\arcsin x)^2} + c$

33. $\int \frac{\cos x}{\sqrt[3]{\sin^2 x}} dx =$

1) $3\sqrt[3]{\sin x} + c$ 2) $3\sqrt[3]{\sin^2 x} + c$

3) $\sqrt[3]{\sin x} + c$ 4) $\sqrt[3]{\sin^2 x} + c$

34. $\int \frac{1}{(\sin^{-1} x)^2 \sqrt{1-x^2}} dx$

1) $\sin^{-1} x + c$ 2) $\frac{1}{\sin^{-1} x} + c$

3) $\frac{-1}{\sin^{-1} x} + c$ 4) $\frac{(\sin^{-1} x)^2}{2} + c$

35. $\int \sec^2 x \cdot (1 - \tan x)^n dx, (n \neq -1) =$

1) $\frac{-(1 - \tan x)^{n+1}}{n+1} + c$ 2) $\frac{(1 - \tan x)^{n+1}}{n+1} + c$

3) $\frac{(1 - \tan x)^n}{n} + c$ 4) $\frac{-(1 - \tan x)^n}{n} + c$

36. $\int \sec^2 x \cdot \sec x \cdot \tan x dx =$

1) $\frac{\sec^3 x}{3} + c$ 2) $-\frac{\sec^3 x}{3} + c$

3) $\frac{\sec^4 x}{11} + c$ 4) $\frac{\sec^4 x}{4} + c$

37. $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx =$

1) $\log |e^x + e^{-x}| + c$ 2) $\log |e^x - e^{-x}| + c$

3) $\frac{1}{2} \log |e^x + e^{-x}| + c$ 4) $\frac{1}{2} \log |e^x - e^{-x}| + c$

38. $\int \frac{1}{(1+x^2)\tan^{-1}x} dx =$

1) $\frac{(\tan^{-1}x)^2}{2} + c$ 2) $\log |\tan^{-1}x| + c$

3) $\log |\sin x + \cos x| + c$ 4) $\log |\sin x - \cos x| + c$

39. $\int \frac{2x+3}{\sqrt{x^2+3x-4}} dx =$

1) $\sqrt{x^2+3x-4} + c$

2) $2\sqrt{x^2+3x-4} + c$

3) $2(x^2+3x-4)^{3/2} + c$

4) $-2(x^2+3x-4)^{3/2} + c$

40. $\int \frac{d(x^2+1)}{\sqrt{x^2+2}}$ is equal to

1) $2\sqrt{x^2+2} + c$

2) $\sqrt{x^2+2} + c$

3) $\frac{1}{(x^2+2)^{3/2}} + c$

4) $-2\sqrt{x^2-2} + c$